Amol Vyas

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LinkedIn | GitHub | Portfolio

Education

RV College of Engineering — Bengaluru, Karnataka, India
Bachelor of Engineering in Computer Science and Engineering — CGPA: 9.39/10 — *Oct 2023 – July 2027*

Experience

Software Engineering Intern — May 2026 – Jul 2026
LinkedIn — *Bengaluru, India*

- Engineered **Agentic AI** solutions within the Productivity Engineering department.
- Designed and implemented the **memory layer** for the AI agent, optimizing context retention and operational efficiency.

Research Intern — Jul 2025 – Jan 2026
Samsung PRISM — *Bengaluru, India*

- Worked on building **deep learning–based models** for detecting **Physical Access Replay Attacks (PA)**.
- Applied the models to **Kannada-language speech** for replay attack detection.

Founding Engineer — Jun 2025 – Jul 2025
Namma Pothole — *Bengaluru, India*

- Built a civic-tech platform consisting of a **WhatsApp bot and website** for Bengaluru citizens to register pothole complaints.
- Developed backend services, WhatsApp bot logic, and frontend interfaces with **zone-wise admin login** for BBMP officials.
- Project was **acknowledged and approved by BBMP** and completed with a team of founding engineers.

Project Intern — Mar 2025 – May 2025
Bhatiyani Astute Institute — *India*

- Developed an **offline NFC-based tap-to-pay wallet** on **ESP32** using **C**, implementing haptic feedback and encryption.

Projects

Hand Gesture Recognition System | *Python, OpenCV, MediaPipe, TensorFlow*

- Developed a hand gesture recognition system enabling **touchless computer control**.
- Implemented gesture, cursor, and **ASL modes** using real-time hand tracking for tasks such as media control and window navigation.
- Optimized the system for accuracy and responsiveness using **MediaPipe** and **OpenCV**.

AI-Powered Legal Document Classification System | *React, FastAPI, LangGraph, RAG, AWS*

- Built an AI-driven platform to digitize and classify legal case documents including digital files, handwritten records, and evidence images.
- Implemented **OCR-based ingestion** via Telegram bots for lawyers, clients, and judges, enabling structured case creation.
- Developed a role-based web portal with separate access for the public, lawyers, and judges, including an **AI chatbot with RAG** for legal research.

Technical Skills

- **Languages:** Python, C, C++, JavaScript, SQL, HTML, CSS
- **Frameworks & Platforms:** LangGraph, FastAPI, React, Express.js
- **AI / ML:** PyTorch, Scikit-learn, YOLOv8, MediaPipe
- **Databases & Storage:** PostgreSQL, MongoDB, AWS S3
- **Developer Tools:** Git, Docker, Linux, AWS, Google Cloud
- **Libraries:** OpenCV, NumPy, pandas, Matplotlib

Achievements & Certifications

- Secured **4th position** at Argonyx '24 (state-level 24-hour hackathon by RV University).
- **Project officially authorized and acknowledged by BBMP** for civic deployment (Namma Pothole).
- Took part in the **Google Cloud Skills Boost** program and earned skill badges in introductory and intermediate **Machine Learning and AI** courses.